

TTE

DESIGN AND ANALYSIS TOOLS FOR TARGET TRIAL EMULATION

A Standard Workflow for Target Trial Emulation with TTE

A practical vignette for TTE version 1.1.1

Inverse probability weighting • Discrete-time survival models • Standardization • Competing risks

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Fully synthetic worked examples. No actual patient records are included.

At a glance

TTE implements the principal standard design and analysis steps required for many target trial emulations with longitudinal observational data (Hernán and Robins, 2016; Hernán et al., 2022, 2026; Cashin et al., 2025). The package keeps each step explicit so that investigators can inspect, diagnose, and report the analysis rather than relying on a single opaque routine.

1 DATA DESIGN check_tte() seqdesign_tte() Check person-period data and construct sequentially nested trials.	2 WEIGHTING est_wt() combine_wt() Estimate treatment, censoring, and adherence weights; then combine them.
3 DIAGNOSTICS balance_wt() diagnose_wt() Assess measured balance, weight distributions, effective sample size, and risk sets.	4 OUTCOME MODELS discsurvreg() Fit weighted pooled discrete-time outcome models with individual-level cluster-robust inference.
5 MARGINAL EFFECTS std_tte() Standardize to survival, risk, cumulative incidence, risk differences, risk ratios, and NNT or NNH.	6 CURVES & UNCERTAINTY curve_tte() boot_tte() Estimate weighted Kaplan-Meier or Aalen-Johansen curves and perform individual-cluster bootstrap.

SCOPE

TTE supplies computational tools. It does not determine whether eligibility criteria, treatment strategies, time zero, outcomes, confounders, or causal assumptions are scientifically appropriate.

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1. Overview

Target trial emulation organizes an observational analysis around the protocol of the randomized trial that would ideally answer the causal question. The protocol defines eligibility criteria, treatment strategies, treatment assignment, time zero, follow-up, outcomes, causal contrasts, and the analysis plan.

TTE provides design and analysis tools for implementing the principal standard steps after these components have been operationalized in longitudinal observational data.

- structural checking of person-period data;
- construction of sequentially nested trials;
- baseline treatment weights;
- longitudinal loss-to-follow-up and adherence weights;
- intention-to-treat and per-protocol analyses;
- covariate-balance and weight diagnostics;
- weighted pooled discrete-time outcome models;
- standardization to marginal risks and treatment contrasts;
- weighted Kaplan-Meier and Aalen-Johansen curves; and
- individual-cluster bootstrap.

DESIGN FIRST

TTE does not infer the target-trial protocol from the data. Eligibility, treatment strategies, time zero, outcome definitions, censoring rules, and estimands must be specified by the investigator before using the package.

2. Installation

Install the CRAN version with:

```
R  
install.packages("TTE")
```

Load the package with:

```
R  
library(TTE)
```

SYNTHETIC DATA

The examples below use the fully synthetic datasets included in the package. They contain no actual patient records, and their numerical results have no clinical interpretation.

3. Package workflow

A typical TTE analysis follows these steps:

1. Specify the target-trial protocol outside the software.
2. Construct or obtain a baseline dataset and person-period dataset.
3. Check the person-period structure.
4. Estimate the baseline treatment weight.
5. Assess covariate balance.
6. Estimate longitudinal censoring or adherence weights.
7. Combine and diagnose the final analysis weight.
8. Fit a weighted discrete-time outcome model.
9. Standardize predictions to marginal risks.
10. Estimate a weighted Kaplan-Meier or Aalen-Johansen curve.

11. Use individual-cluster bootstrap when complete uncertainty propagation is required.

Step	Output
Protocol operationalized	Baseline and person-period data
check_tte() / seqdesign_tte()	Structurally valid person-trial data
est_wt() / combine_wt()	Final ITT or PP analysis weight
balance_wt() / diagnose_wt()	Balance, support, ESS, risk-set diagnostics
discsurvreg()	Weighted relative effect model
std_tte() / curve_tte()	Marginal risks, contrasts, and weighted curves
boot_tte()	Uncertainty propagated at the individual level

4. Data conventions

TTE distinguishes between baseline and person-period data.

A baseline dataset contains one row per trial entry. In a simple active-comparator new-user study, this will often be one row per individual. In a sequence of nested trials, the same individual may contribute multiple trial entries.

A person-period dataset contains one row for every follow-up interval in which a person-trial remains at risk. Typical variables include:

- the original individual identifier;
- the trial identifier;
- the zero-based interval index;
- baseline treatment assignment;
- interval-specific outcomes;
- loss-to-follow-up indicators;
- adherence indicators;
- baseline and time-varying covariates; and
- analysis weights.

CLUSTERING UNIT

The independent unit for variance estimation and bootstrap is generally the original individual, not the person-period row or person-trial entry.

5. Example 1: SGLT2i versus DPP-4i initiation

The SGLT2 example represents an active-comparator new-user target trial emulation among older adults with type 2 diabetes (Noma et al., 2026a). The strategies are initiation of a DPP-4 inhibitor and initiation of an SGLT2 inhibitor. The outcome is all-cause death.

5.1. Load and inspect the data

Load the baseline and person-period datasets:

```
R
data(SGLT2_baseline)
data(SGLT2)

baseline <- SGLT2_baseline
long <- SGLT2
```

The baseline dataset contains one row per treatment initiator. The long dataset contains person-month records.

```
R
dim(baseline)
dim(long)

table(baseline$treatment)
with(long, tapply(Y_death, treatment, sum))
```

5.2. Check the person-period structure

Before estimating weights or fitting outcome models, check the data structure.

```
R
check <- check_tte(
  long,
  id = id,
  trial = trial,
  time = time,
  event = Y_death,
  treatment = A
)

print(check)
```

The function detects duplicated intervals, incorrect ordering, gaps in interval indices, post-event rows, within-trial changes in baseline treatment, and invalid weights.

INTERPRETATION

Passing these checks does not establish causal validity. It only indicates that the dataset is structurally compatible with the specified analysis.

5.3. Estimate the baseline treatment weight

The baseline stabilized treatment weight is:

$$SW_i^A = Pr(A_i = a_i) / Pr(A_i = a_i | L_i)$$

By default, `est_wt()` constructs stabilized weights. For baseline treatment weights, the numerator is the marginal probability of the treatment actually received.

```
R
w_treatment <- est_wt(
  A ~
    age + I(age^2) +
    female +
    bmi +
    hba1c + I(hba1c^2) +
    egfr +
    proteinuria +
    prior_heart_failure +
    prior_stroke +
    recent_hospitalization +
    trial_period,
  data = baseline,
  type = "treatment",
  stabilize = TRUE,
  truncate = c(0, 1)
)

summary(w_treatment)
plot(w_treatment)
```

The component is left untruncated because truncation will be applied after the treatment and censoring weights have been combined.

Intermediate quantities can be inspected with:

```
R
head(as.data.frame(w_treatment))
```

The returned data include the observed numerator and denominator probabilities, the interval factor, the untruncated weight, and the final weight.

5.4. Assess baseline balance

```
R
balance <- balance_wt(
  A ~
    age +
    female +
    bmi +
    hba1c +
    egfr +
    proteinuria +
    prior_heart_failure +
    prior_stroke +
    recent_hospitalization +
    trial_period,
  data = baseline,
  weights = w_treatment
)

print(balance)
summary(balance, threshold = 0.10)
plot(balance, threshold = 0.10)
```

DIAGNOSTIC THRESHOLD

An absolute standardized mean difference of 0.10 is often used as a diagnostic threshold. This convention does not demonstrate that exchangeability holds or that unmeasured confounding is absent.

5.5. Map the baseline weight to person-period data

The treatment weight is estimated once for each baseline trial entry and must then be copied to all corresponding person-period rows.

```
R
key_baseline <- paste(
  baseline$id,
  baseline$trial,
  sep = ":"
)

key_long <- paste(
  long$id,
  long$trial,
  sep = ":"
)

baseline_index <- match(key_long, key_baseline)

stopifnot(!anyNA(baseline_index))

long$w_treatment_est <- weights(
  w_treatment,
  which = "untruncated"
)[baseline_index]
```

Matching by both individual and trial makes the same code applicable to sequentially nested trials.

5.6. Estimate the loss-to-follow-up weight

The indicator `stay_ltfu` equals 1 when the person remains observed through the next interval.

```
R
w_ltfu <- est_wt(
  stay_ltfu ~
  A +
```

```

splines::ns(time, df = 3) +
age +
female +
bmi +
hba1c +
egfr +
proteinuria +
prior_heart_failure +
prior_stroke +
recent_hospitalization +
trial_period,
numerator =
stay_ltfu ~
A +
splines::ns(time, df = 3) +
trial_period,
data = long,
type = "censoring",
id = id,
trial = trial,
time = time,
cumulative = TRUE,
lag = 1,
stabilize = TRUE,
truncate = c(0, 1)
)

summary(w_ltfu)
head(as.data.frame(w_ltfu))

```

TEMPORAL ALIGNMENT

With `cumulative = TRUE`, interval-specific factors are multiplied within each person-trial. With `lag = 1`, the outcome contribution in an interval is weighted using information available through the start of that interval.

5.7. Construct and diagnose the final ITT weight

```

R
w_itt <- combine_wt(
  long$w_treatment_est,
  w_ltfu,
  truncate = c(0.01, 0.99),
  normalize = "none"
)

long$w_itt_est <- weights(w_itt)

```

The final weight is the product of the baseline treatment weight and cumulative loss-to-follow-up weight, truncated at the 1st and 99th percentiles.

```

R
diagnostics <- diagnose_wt(
  w_itt_est,
  data = long,
  treatment = treatment,
  time = time,
  id = id,
  trial = trial
)

print(diagnostics)

old_par <- par(no.readonly = TRUE)
par(mfrow = c(1, 2))

plot(diagnostics, type = "weights")
plot(diagnostics, type = "risk_set")

par(old_par)

```

EFFECTIVE INFORMATION

The risk-set plot displays the effective rather than nominal risk-set size. Late follow-up may contain little effective information even when several person-period rows remain.

5.8. Fit the weighted discrete-time outcome model

```
R
fit_death <- discsurvreg(
  Y_death ~
    A +
    splines::ns(time, df = 3) +
    trial_period,
  data = long,
  id = id,
  weights = w_itt_est,
  family = quasibinomial(link = "cloglog"),
  var_method = "standard"
)

print(fit_death)
summary(fit_death)

confint(
  fit_death,
  parm = "A",
  eform = TRUE
)
```

With the complementary log-log link, the exponentiated treatment coefficient is a discrete-time hazard ratio under the fitted model. The cluster-robust covariance is calculated using the original individual identifier.

5.9. Standardize to marginal risks

```
R
std_death <- std_tte(
  fit_death,
  data = baseline,
  treatment = A,
  time = time,
  times = 0:59,
  values = c(0, 1),
  labels = c("DPP-4i", "SGLT2i")
)

summary(std_death, horizon = 36)
summary(std_death, horizon = 60)

plot(std_death, measure = "risk")
```

The function creates counterfactual copies of the baseline population under each treatment strategy and averages the predicted risks.

- strategy-specific survival;
- strategy-specific cumulative risk;
- risk difference;
- risk ratio;
- survival difference; and
- time-specific NNT or NNH.

TIME CONVENTION

The values 0 through 59 represent 60 follow-up intervals. Public summaries use elapsed time, so horizon = 60 reports the result after 60 intervals.

5.10. Estimate a weighted Kaplan-Meier curve

```
R
km_death <- curve_tte(
  long,
  time = time,
  event = Y_death,
  treatment = treatment,
  weights = w_itt_est,
  type = "km",
  id = id,
  trial = trial
)

summary(km_death, time = 36)
summary(km_death, time = 60)

plot(km_death, measure = "risk")
```

The standardized curve is model based. The weighted Kaplan-Meier curve is estimated directly from weighted risk sets.

COMPARE ESTIMATORS

The two curves need not be identical. Large differences should prompt examination of the outcome model, weight models, positivity, and effective risk sets.

6. Per-protocol analysis

The intention-to-treat estimand compares baseline initiation strategies regardless of later switching or discontinuation.

The per-protocol estimand considers sustained compatibility with the assigned strategy. Follow-up is artificially censored at protocol deviation, and inverse probability weighting addresses the resulting selection.

Retain the intervals in the per-protocol risk set:

```
R
pp_data <- subset(
  long,
  pp_at_risk == 1
)

pp_data$stay_pp <- as.integer(
  pp_data$stay_ltfu == 1 &
  pp_data$stay_adherent == 1
)
```

Estimate a joint censoring weight for remaining observed and adherent:

```
R
w_pp_censoring <- est_wt(
  stay_pp ~
    A +
    splines::ns(time, df = 3) +
    age +
    female +
    bmi +
    hba1c +
    egfr +
    proteinuria +
    prior_heart_failure +
    prior_stroke +
    recent_hospitalization +
    trial_period,
  numerator =
    stay_pp ~
    A +
    splines::ns(time, df = 3) +
    trial_period,
  data = pp_data,
```

```

type = "censoring",
id = id,
trial = trial,
time = time,
cumulative = TRUE,
lag = 1,
stabilize = TRUE,
truncate = c(0, 1)
)

```

Construct the final PP weight:

```

R
w_pp <- combine_wt(
  pp_data$w_treatment_est,
  w_pp_censoring,
  truncate = c(0.01, 0.99),
  normalize = "none"
)

pp_data$w_pp_est <- weights(w_pp)

```

The PP outcome model and standardized risks are then obtained using the same functions as in the ITT analysis.

```

R
fit_death_pp <- discsurvreg(
  Y_death ~
    A +
    splines::ns(time, df = 3) +
    trial_period,
  data = pp_data,
  id = id,
  weights = w_pp_est,
  family = quasibinomial(link = "cloglog"),
  var_method = "standard"
)

std_death_pp <- std_tte(
  fit_death_pp,
  data = baseline,
  treatment = A,
  time = time,
  times = 0:59,
  values = c(0, 1),
  labels = c("DPP-4i", "SGLT2i")
)

```

SUBSTANTIVE REQUIREMENT

A valid per-protocol analysis requires measurement of the time-varying predictors of adherence and outcome. The synthetic SGLT2 dataset contains limited updated clinical information, so this example primarily demonstrates the computational structure.

7. Example 2: sequential trials and competing risks

The ARB example represents sequentially nested trials comparing ARB and CCB strategies in chronic kidney disease (Noma et al., 2026b). Heart-failure hospitalization is the event of interest, and death is a competing event.

```

R
data(ARB_baseline)
data(ARB)

arb_baseline <- ARB_baseline
arb_long <- ARB

```

The baseline dataset contains 900 person-trial entries contributed by 750 original individuals. The same person can enter more than one trial. All matching, cumulative-weight calculation, variance estimation, and bootstrap procedures must therefore retain both the trial identifier and original individual identifier.

7.1. Cause-specific outcome models

Fit one model for heart failure:

```
R
fit_hf <- discsurvreg(
  Y_hf ~
    A +
    splines::ns(time, df = 3) +
    trial_period,
  data = arb_long,
  id = id,
  weights = w_itt_est,
  family = quasibinomial(link = "cloglog"),
  var_method = "standard"
)
```

Fit a second model for competing death:

```
R
fit_death <- discsurvreg(
  Y_death ~
    A +
    splines::ns(time, df = 3) +
    trial_period,
  data = arb_long,
  id = id,
  weights = w_itt_est,
  family = quasibinomial(link = "cloglog"),
  var_method = "standard"
)
```

MODEL INTERPRETATION

The treatment coefficients are cause-specific discrete-time hazard ratios. They are not Fine-Gray subdistribution hazard ratios.

ABBREVIATED CODE

This section assumes that `w_itt_est` has already been estimated from `ARB_baseline` and `ARB` using the same treatment-weight, censoring-weight, combination, and diagnostic steps shown in Example 1. The complete executable ARB tutorial is available in the public repository.

7.2. Standardized cumulative incidence

Supply both cause-specific models to `std_tte()`:

```
R
std_hf <- std_tte(
  fit_hf,
  data = arb_baseline,
  treatment = A,
  time = time,
  times = 0:59,
  competing_fit = fit_death,
  values = c(0, 1),
  labels = c("CCB", "ARB")
)

summary(std_hf, horizon = 36)
summary(std_hf, horizon = 60)

plot(std_hf, measure = "risk")
```

The function combines the predicted heart-failure and death probabilities to estimate the cumulative incidence of heart-failure hospitalization.

COMPETING RISK

One minus Kaplan-Meier should not be used for this outcome because death precludes subsequent heart-failure hospitalization.

7.3. Weighted Aalen-Johansen curve

The person-period event code is:

- 0: no event;
- 1: heart-failure hospitalization;
- 2: death.

The weighted cumulative incidence is estimated with:

```
R
aj_hf <- curve_tte(
  arb_long,
  time = time,
  event = event_code,
  treatment = treatment,
  weights = w_itt_est,
  type = "aj",
  cause = 1,
  id = id,
  trial = trial
)

summary(aj_hf, time = 60)
plot(aj_hf, measure = "risk")
```

The standardized cumulative-incidence curve is model based. The weighted Aalen-Johansen curve is calculated directly from weighted cause-specific risk sets.

8. Sequentially nested trial construction

The function `seqdesign_tte()` creates trial entries from longitudinal source data.

```
R
expanded <- seqdesign_tte(
  source_data,
  id = id,
  calendar_time = month,
  eligible = eligible,
  treatment = treatment,
  outcome = outcome,
  censor = censor,
  max_follow = 60,
  induction = 1
)
```

At each eligible treatment-decision time, a new trial entry is created. The person may enter again at another calendar time if eligibility is satisfied.

PROTOCOL RESPONSIBILITY

The function does not determine whether the eligibility criteria, washout period, or treatment strategies are scientifically appropriate. These rules must be encoded before expansion.

9. Bootstrap inference

The function `boot_tte()` resamples original individuals.

```

R
statistic <- function(d) {
  # Re-estimate treatment and censoring models here.
  # Reconstruct the final weight.
  # Refit the outcome model.
  # Return the desired estimates as a named numeric vector.
}

bootstrap_result <- boot_tte(
  data = long_data,
  id = id,
  statistic = statistic,
  R = 500,
  conf_level = 0.95,
  seed = 2026
)

```

For complete uncertainty propagation, the statistic function should repeat the entire analysis pipeline inside each bootstrap sample.

SAMPLING UNIT

The bootstrap must sample original individuals rather than individual person-period rows. This retains all periods and sequential-trial entries belonging to the sampled person.

10. Recommended reporting

A target trial emulation using TTE should report:

- the complete target-trial protocol;
- the operational definition of each protocol component;
- the baseline and longitudinal data structures;
- numerator and denominator weight models;
- the temporal ordering and lag used for longitudinal weights;
- truncation and normalization rules;
- covariate-balance diagnostics;
- weight distributions and effective sample sizes;
- effective risk sets over follow-up;
- the outcome-model link and time function;
- the population used for standardization;
- relative and absolute effect estimates;
- competing-risk handling;
- bootstrap cluster definition; and
- sensitivity analyses addressing model and design choices.

REPORTING GUIDELINE

The TARGET Statement should be used when preparing the study report.

11. Limitations

TTE version 1.1.1 focuses on common analyses involving static treatment strategies and inverse probability weighting.

It does not provide general-purpose tools for:

- dynamic treatment regimes;
- complex cloning and grace-period strategies;
- continuous treatments;
- targeted maximum likelihood estimation;
- automatic machine-learning nuisance models;
- joint calibrated weighting;

- integrated multiple imputation; or
- distributed or database-backed computation for extremely large datasets.

SCIENTIFIC VALIDITY

Software cannot resolve violations of exchangeability, consistency, positivity, or measurement validity. The target population and intervention strategies must remain clinically meaningful.

12. Further examples and citation

Complete executable scripts are available for:

- SGLT2i versus DPP-4i initiation with all-cause death; and
- ARB versus CCB strategies with heart-failure hospitalization and competing death.

Each example is available from the public TTE repository.

<https://github.com/nomahi/TTE>

After installation, obtain the package citation with:

```
R
citation("TTE")
```

Record the analysis environment at the end of each analysis:

```
R
sessionInfo()
```

The software version, complete code, weight-model formulas, random seeds, and package dependencies should be archived with the analysis.

Appendix A. Function map

Function	Purpose	Principal output
check_tte()	Check person-period data	Duplicates, timing, post-event rows, treatment changes, invalid weights
seqdesign_tte()	Construct sequential trials	Stacked person-trial data with interval follow-up
est_wt()	Estimate inverse probability weights	Treatment, censoring, or adherence weights with retained probabilities and factors
combine_wt()	Combine weight components	Final truncated or normalized analysis weight
balance_wt()	Assess measured balance	Unweighted and weighted standardized mean differences
diagnose_wt()	Diagnose weights	Distribution, ESS, treatment-specific behavior, effective risk sets
discsurvreg()	Fit outcome model	Weighted pooled discrete-time model with cluster-robust covariance
std_tte()	Obtain marginal effects	Survival, risk, CIF, RD, RR, NNT or NNH
curve_tte()	Estimate weighted curves	Right-continuous Kaplan-Meier or Aalen-Johansen curves
boot_tte()	Bootstrap original individuals	Replicates and percentile intervals preserving repeated records

References

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